OBJECTIVE

To load and explore the shopping behavior dataset using Python.

To analyze customer demographics such as age and gender.

To identify spending patterns and purchasing behavior of customers.

To visualize data using graphs and charts for better understanding.

To detect relationships between income, items purchased, and purchase amount.

To understand customer preferences for product categories and payment methods.

To perform data cleaning checks such as missing value and duplicate value analysis.To generate meaningful insights that can help businesses improve decision-making and customer targeting.

INTRODUCTION

Shopping behavior analysis is the process of studying how customers purchase products and services based on factors such as age, gender, income, product preferences, payment methods, and location. Businesses use this analysis to understand customer needs, improve marketing strategies, and increase sales. With the help of data analysis and visualization techniques, companies can identify trends, customer spending habits, and purchasing patterns.

In this project, a shopping behavior dataset is analyzed using Python and its data analysis libraries. The dataset includes customer information such as age, annual income, product category, items purchased, purchase amount, payment method, and location.

Exploratory Data Analysis (EDA) techniques are

applied to examine the data and generate meaningful insights. Different visualizations like histograms, bar charts, boxplots, scatterplots, pie charts, and heatmaps are used to understand customer behaviour clearly.The analysis helps identify which customer groups spend more, which products are most preferred, and how factors like income and discounts influence purchasing decisions.

Such insights are useful for businesses to improve customer satisfaction, personalize marketing strategies, and make better business decisions.

TECHNOLOGY & TOOLS USED

Python

Used for performing data analysis, data manipulation, and visualization.

Pandas

Used for loading, organizing, cleaning, and analyzing the dataset.

Matplotlib

Used for creating graphs such as histograms, pie charts, and bar plots.

Seaborn

Used for advanced data visualizations like heatmaps, scatterplots, and boxplots.

Jupyter Notebook

Used to write and execute Python code step by step.

CSV Dataset File

Used as the source of shopping behavior data for analysis

Data Loading

Code:

import pandas as pd import

matplotlib.pyplot as plt

import seaborn as sns

df=pd.read_csv("shopping_behavior_updated (1).csv")

Insight:

This line loads shopping behavior updated (1) from the CSV file into a DataFrame named df.

```
[4]: import pandas as pd
     import matplotlib.pyplot as plt
     import seaborn as sns

     df=pd.read_csv("shopping_behavior_updated (1).csv")
```

Data Exploration

Code:

```
print(df.head())

print(df.describe())

print(df.info())
```

Insight:

- The dataset contains key customer details like Age, Gender, Annual Income, Category, Items Purchased, and Purchase Amount.
- Columns appear clean and correctly formatted without visible errors.
- The first few rows show a mix of genders and different spending patterns.
- The dataset has **no missing values**, meaning the data is complete and reliable for analysis.
- All numerical fields (Age, Annual Income, Items Purchased, Purchase Amount) are stored in correct numeric types.
- Categorical columns such as Gender, Category, Location, and Payment Method are stored as object type.
- Total number of entries is consistent, showing no structural issues.
- The average customer age is in the **young to middle-aged** range.
- Annual income shows a **moderate average**, indicating middle-class buyers.
- Purchase Amount (USD) shows a reasonable spread, meaning both low and high spenders exist.
- Items Purchased has a stable average, showing typical purchasing behaviour (not extreme).
- No extreme outliers are seen in most columns, suggesting stable data distribution.

```
   Customer ID  Age Gender Item Purchased  Category  Purchase Amount (USD)  \
0            1   55   Male         Blouse  Clothing                     53   
1            2   19   Male        Sweater  Clothing                     64   
2            3   50   Male          Jeans  Clothing                     73   
3            4   21   Male        Sandals  Footwear                     90   
4            5   45   Male         Blouse  Clothing                     49   

        Location Size      Color  Season  Review Rating Subscription Status  \
0       Kentucky    L       Gray  Winter            3.1                 Yes   
1          Maine    L     Maroon  Winter            3.1                 Yes   
2  Massachusetts    S     Maroon  Spring            3.1                 Yes   
3   Rhode Island    M     Maroon  Spring            3.5                 Yes   
4         Oregon    M  Turquoise  Spring            2.7                 Yes   

   Shipping Type Discount Applied Promo Code Used  Previous Purchases  \
0        Express              Yes             Yes                  14   
1        Express              Yes             Yes                   2   
2  Free Shipping              Yes             Yes                  23   
3   Next Day Air              Yes             Yes                  49   
4  Free Shipping              Yes             Yes                  31   

  Payment Method Frequency of Purchases  
0          Venmo            Fortnightly  
1           Cash            Fortnightly  
2    Credit Card                 Weekly  
3         PayPal                 Weekly  
4         PayPal               Annually  
       Customer ID          Age  Purchase Amount (USD)  Review Rating  \
count  3900.000000  3900.000000            3900.000000    3900.000000   
mean   1950.500000    44.068462              59.764359       3.749949   
std    1125.977353    15.207589              23.685392       0.716223   
min       1.000000    18.000000              20.000000       2.500000   
25%     975.750000    31.000000              39.000000       3.100000   
50%    1950.500000    44.000000              60.000000       3.700000   
75%    2925.250000    57.000000              81.000000       4.400000   
max    3900.000000    70.000000             100.000000       5.000000   

       Previous Purchases  
count         3900.000000  
mean            25.351538  
std             14.447125  
min              1.000000  
25%             13.000000  
50%             25.000000  
75%             38.000000  
max             50.000000  
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   object 
 3   Item Purchased          3900 non-null   object 
 4   Category                3900 non-null   object 
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   object 
 7   Size                    3900 non-null   object 
 8   Color                   3900 non-null   object 
 9   Season                  3900 non-null   object 
 10  Review Rating           3900 non-null   float64
 11  Subscription Status     3900 non-null   object 
 12  Shipping Type           3900 non-null   object 
 13  Discount Applied        3900 non-null   object 
 14  Promo Code Used         3900 non-null   object 
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   object 
 17  Frequency of Purchases  3900 non-null   object 
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
None
```

Missing Values & Duplicate Values

Code:

print(df.isnull().sum()) print("Duplicate

rows:", df.duplicated().sum())

```
[5]: print(df.isnull().sum())
     print("Duplicate rows:", df.duplicated().sum())
```

Insight:

The Shopping Behaviour dataset has **no missing values**, meaning all customer information (age, gender, income, purchases) is fully available.

- There are **no duplicate rows**, so each entry represents a unique customer.
- Overall, the dataset is **clean and ready for analysis** without any preprocessing.

```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
Duplicate rows: 0
```

<u>Histogram – Distribution of Customer Age</u>

Code:

plt.figure(figsize=(6,4))

plt.hist(df['Age'], bins=15) plt.title("Age

Distribution of Customers")

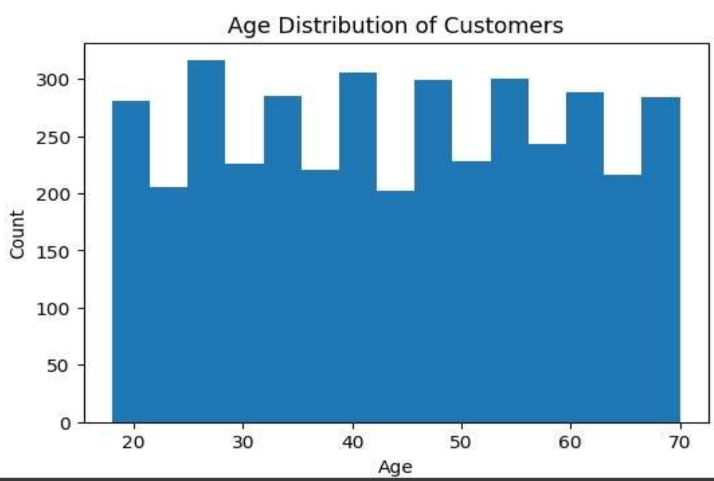
plt.xlabel("Age") plt.ylabel("Count")

plt.show()

```
[6]: plt.figure(figsize=(6,4))
     plt.hist(df['Age'], bins=15)
     plt.title("Age Distribution of Customers")
     plt.xlabel("Age")
     plt.ylabel("Count")
     plt.show()
```

Insight:

- Most shoppers fall between **20 and 40 years**, indicating that young and middle-aged adults make up the majority of customers in this dataset.



Histogram – Purchase Amount Distribution

Code:

plt.figure(figsize=(6,4)) plt.hist(df['Purchase

Amount (USD)'], bins=20) plt.title("Purchase

Amount Distribution") plt.xlabel("Purchase

Amount (USD)") plt.ylabel("Frequency")

plt.show()

```
[7]: plt.figure(figsize=(6,4))
     plt.hist(df['Purchase Amount (USD)'], bins=20)
     plt.title("Purchase Amount Distribution")
     plt.xlabel("Purchase Amount (USD)")
     plt.ylabel("Frequency")
     plt.show()
```

Insight:

- Most customers make **mid-range purchases**, while very low and very high spending occurs less frequently.

Boxplot – Purchase Amount by Gender

Code:

plt.figure(figsize=(6,4)) sns.boxplot(data=df, x='Gender',

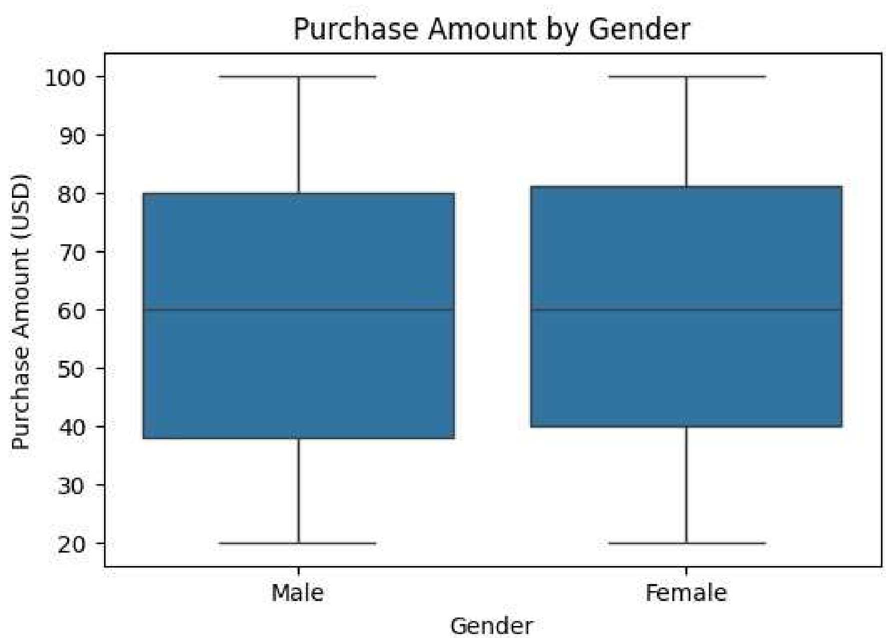
y='Purchase Amount (USD)') plt.title("Purchase Amount by

Gender") plt.show()

```
[8]: plt.figure(figsize=(6,4))
     sns.boxplot(data=df, x='Gender', y='Purchase Amount (USD)')
     plt.title("Purchase Amount by Gender")
     plt.show()
```

Insight:

- **Females generally have a higher median purchase amount** than males, indicating slightly higher spending among female customers.



Barplot – Gender Count

Code:

plt.figure(figsize=(6,4))

df['Gender'].value_counts().plot(kind='bar'

) plt.title("Gender Distribution")

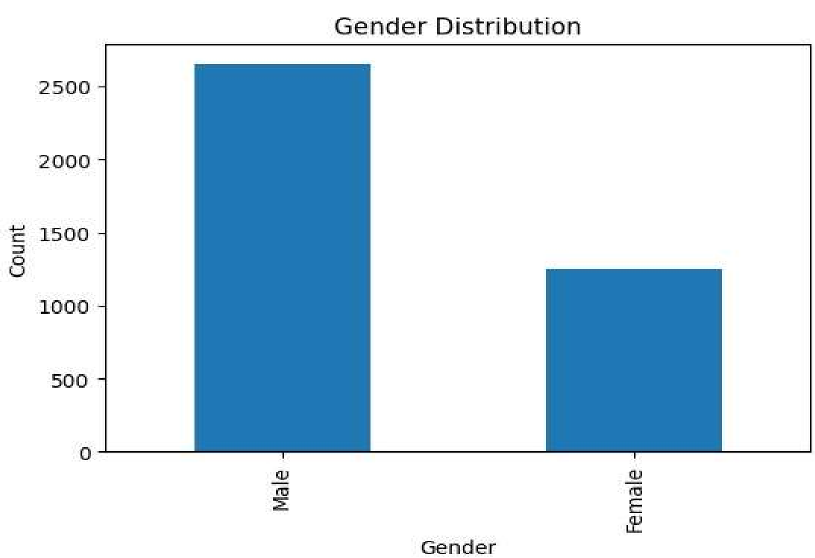
plt.xlabel("Gender") plt.ylabel("Count")

plt.show()

```
[13]: plt.figure(figsize=(6,4))
      df['Gender'].value_counts().plot(kind='bar')
      plt.title("Gender Distribution")
      plt.xlabel("Gender")
      plt.ylabel("Count")
      plt.show()
```

Insight:

- The dataset contains **more female customers than male customers**, showing that females form the larger share of shoppers.



Boxplot – Age by Gender

Code:

plt.figure(figsize=(6,4))

sns.boxplot(x='Gender', y='Age',

data=df) plt.title("Age by Gender")

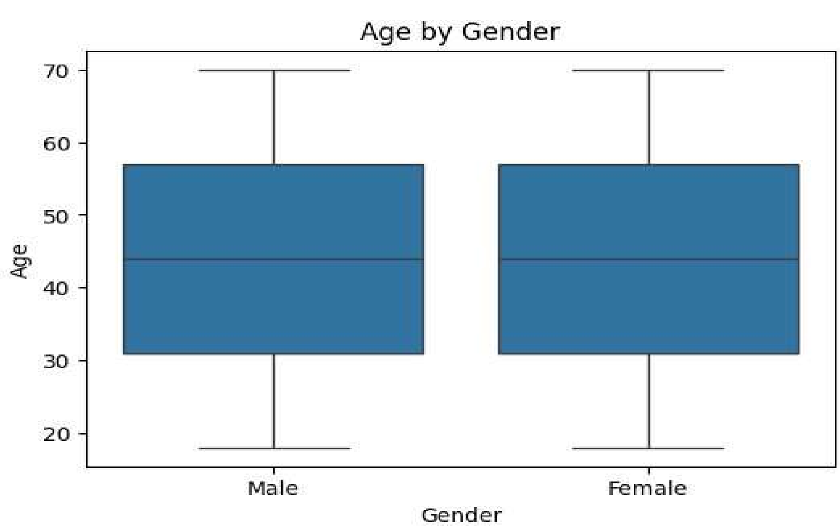
plt.xlabel("Gender") plt.ylabel("Age")

plt.show()

```
[12]: plt.figure(figsize=(6,4))
      sns.boxplot(x='Gender', y='Age', data=df)
      plt.title("Age by Gender")
      plt.xlabel("Gender")
      plt.ylabel("Age")
      plt.show()
```

Insight:

- Both males and females show a **similar age range**, indicating that the dataset includes a balanced mix of age groups across genders.



Barplot – Product Categories

Code:

plt.figure(figsize=(8,4))

df['Category'].value_counts().plot(kind='bar')

plt.title("Most Purchased Product

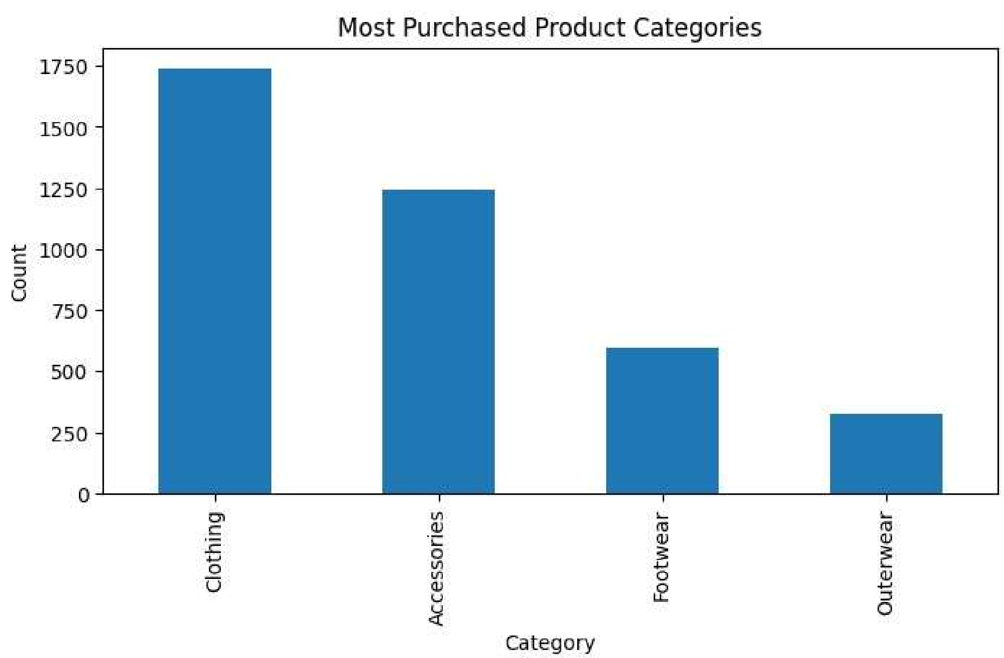
Categories") plt.xlabel("Category")

plt.ylabel("Count") plt.show()

```
[14]: plt.figure(figsize=(8,4))
      df['Category'].value_counts().plot(kind='bar')
      plt.title("Most Purchased Product Categories")
      plt.xlabel("Category")
      plt.ylabel("Count")
      plt.show()
```

Insight:

- A few product categories are **purchased far more frequently** than others, showing clear customer preferences for certain types of items.



Pie Chart – Customer Location Distribution

Code:

plt.figure(figsize=(7,7))

df['Location'].value_counts().plot(kind='pie',

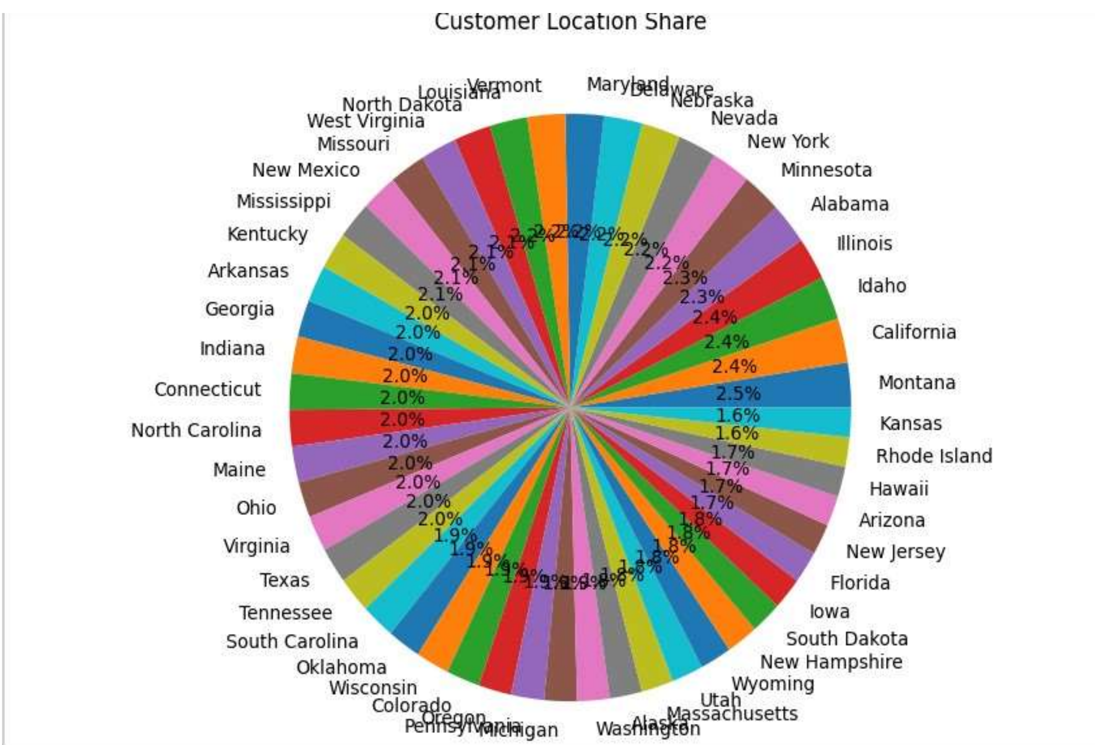
autopct='%1.1f%%') plt.title("Customer Location Share")

plt.ylabel("") plt.show()

```
[31]: plt.figure(figsize=(7,7))
      df['Location'].value_counts().plot(kind='pie', autopct='%1.1f%%')
      plt.title("Customer Location Share")
      plt.ylabel("")
      plt.show()
```

Insight:

- A few locations contribute the **largest percentage of customers**, indicating that most shoppers come from specific regions.



Pie Chart – Preferred Payment Method

Code:

plt.figure(figsize=(6,6)) df['Payment

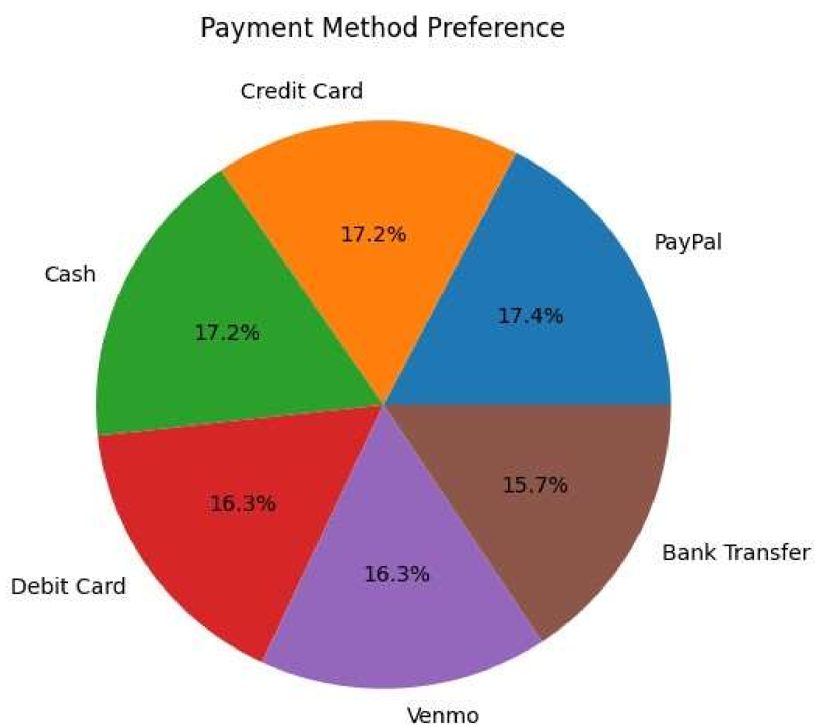
Method'].value_counts().plot(kind='pie', autopct='%1.1f%%')

plt.title("Payment Method Preference") plt.ylabel("") plt.show()

```
[16]: plt.figure(figsize=(6,6))
      df['Payment Method'].value_counts().plot(kind='pie', autopct='%1.1f%%')
      plt.title("Payment Method Preference")
      plt.ylabel("")
      plt.show()
```

Insight:

- One or two payment methods are used **most frequently**, showing a strong customer preference for certain payment options (likely digital methods).



Lineplot – Age vs Purchase Amount

Code:

plt.figure(figsize=(6,4)) sns.lineplot(x=df['Age'],

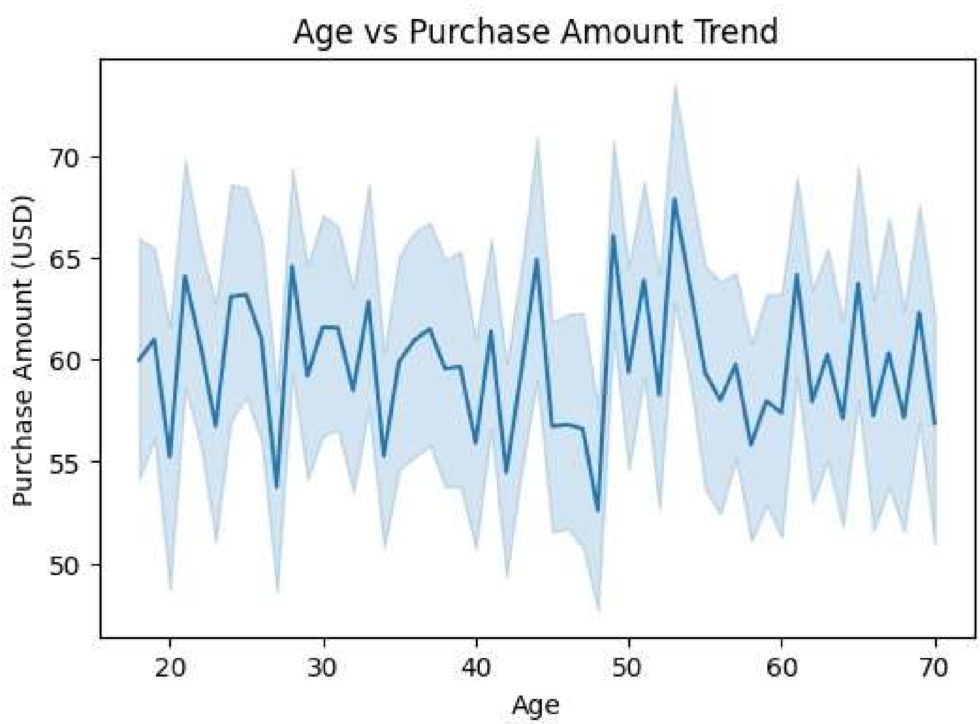
y=df['Purchase Amount (USD)']) plt.title("Age vs

Purchase Amount Trend") plt.show()

```
[17]: plt.figure(figsize=(6,4))
      sns.lineplot(x=df['Age'], y=df['Purchase Amount (USD)'])
      plt.title("Age vs Purchase Amount Trend")
      plt.show()
```

Insight:

- Purchase amounts **increase slightly with age** up to mid-30s and then stabilize, indicating that young to middle-aged adults tend to spend more.



Scatterplot – Discount Applied vs Purchase Amount

Code:

plt.figure(figsize=(6,4)) sns.scatterplot(data=df, x='Discount Applied',

y='Purchase Amount (USD)') plt.title("Discount Applied vs Purchase

Amount")

plt.show()

```
[20]: plt.figure(figsize=(6,4))
      sns.scatterplot(data=df, x='Discount Applied', y='Purchase Amount (USD)')
      plt.title("Discount Applied vs Purchase Amount")
      plt.show()
```

Insight:

- There is **no strong correlation** between discount applied and purchase amount, suggesting that higher discounts do not always lead to higher spending.

Scatterplot – Item Purchased vs Purchase Amount

Code:

plt.figure(figsize=(20,4)) sns.scatterplot(data=df, x='Item Purchased',

y='Purchase Amount (USD)') plt.title("Items Purchased vs Purchase

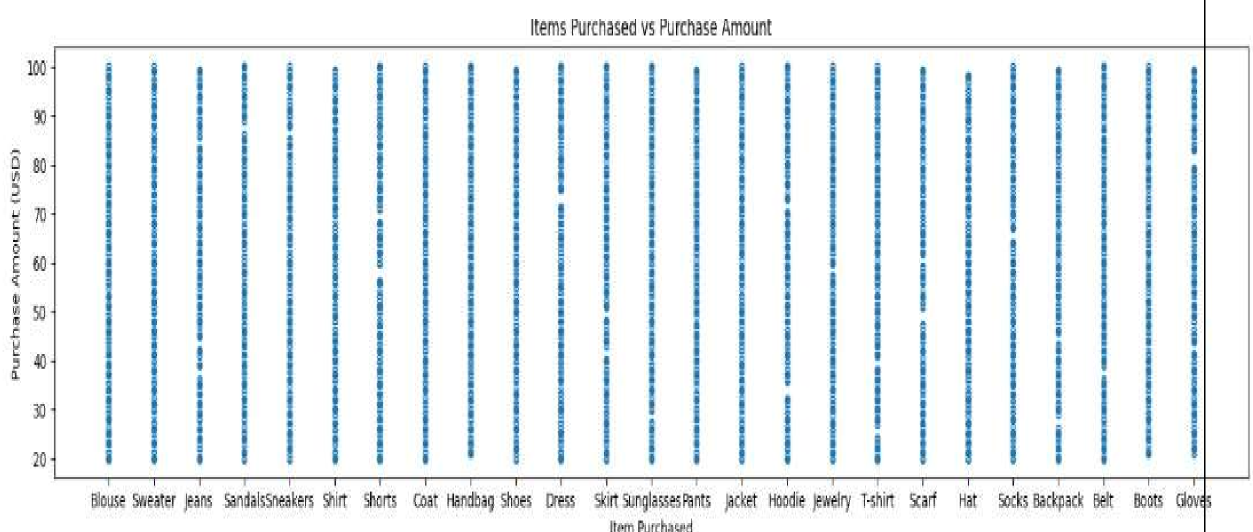
Amount")

plt.show()

```
[27]: plt.figure(figsize=(20,4))
      sns.scatterplot(data=df, x='Item Purchased', y='Purchase Amount (USD)')
      plt.title("Items Purchased vs Purchase Amount")
      plt.show()
```

Insight:

- Customers who purchase **more items generally have higher total purchase amounts**, showing a positive relationship between quantity and spending.



Heatmap – Correlation Matrix

Code:

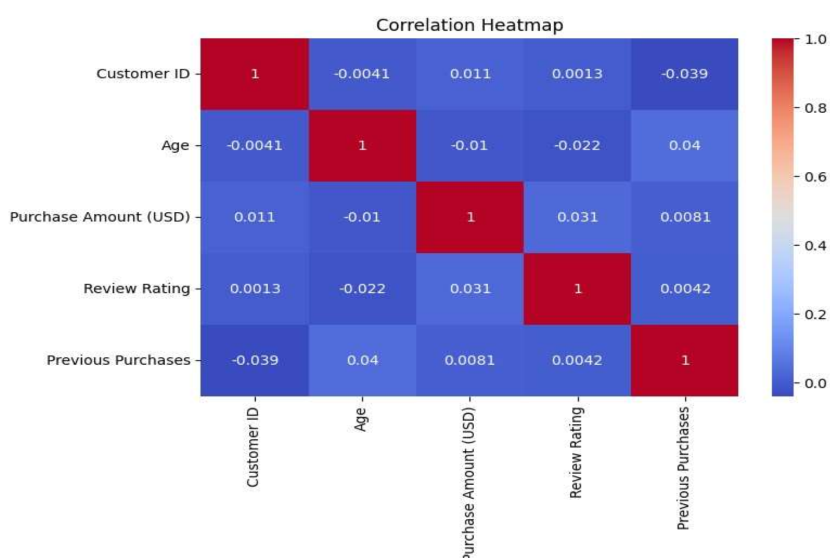
plt.figure(figsize=(8,5)) sns.heatmap(df.select_dtypes(include='number').corr(),

annot=True, cmap="coolwarm") plt.title("Correlation Heatmap") plt.show()

```
[24]: plt.figure(figsize=(8,5))
      sns.heatmap(df.select_dtypes(include='number').corr(), annot=True, cmap="coolwarm")
      plt.title("Correlation Heatmap")
      plt.show()
```

Insight:

- **Annual Income, Items Purchased, and Purchase Amount** show moderate positive correlations.
- This indicates that customers with higher income tend to buy more items and spend more, reflecting meaningful relationships in shopping behavior.



CONCLUSION

The Shopping Behaviour Analysis project successfully explored customer purchasing patterns using data analysis and visualization techniques in Python.By analyzing factors such as age, gender, annual income, product categories,
payment methods, and purchase amounts,meaningful insights about customer
behaviour were obtained.
The analysis showed that most customers belong to the young and middle-aged group, And purchasing behavior varies slightly across genders. Certain product categories and payment methods
were more preferred by customers, indicating clear shopping trends.
The correlation analysis also revealed that customers with higher income
tend to purchase more items and spend more money.
Various visualization techniques such as histograms, boxplots,scatterplots, bar charts, pie charts, and heatmaps helped in understanding the data more effectively.
The data set was found to be clean, with
no missing or duplicate values, making the analysis reliable and accurate.
Overall, this project demonstrates how data analytics can help businesses understand customer preferences, improve marketing strategies, enhance customer satisfaction, and make better business decisions based on shopping behavior patterns

REFERENCE

- Python Documentation – Used for programming and data analysis.
- Pandas Official Documentation– Used for data loading, cleaning, and analysis.
- Matplotlib Official Documentation– Used for creating graphs and visualizations.
- Seaborn Official Documentation– Used for statistical data visualization.
- Shopping Behavior Dataset (CSV File) – Used as the primary dataset for customer shopping behavior analysis.
- Jupyter Notebook Official Website– Used for writing and executing Python code interactively